

Alpecca feature & function skeleton (tiered)

Last reviewed 2026-07-09 - regenerated from multi-subagent code audit. Canonical stack: PROJECT_CONTEXT.md, HANDOFF.md, docs/AGENTIC_ASSESSMENT.md, docs/ALPECCA_CURRENT_PROGRESS.md. Legend: green done, amber partial, gray not started, blue parked, slate superseded.

Tier 1: Foundation runtime

DONE	Fast chat path + runtime status + stream routing Draft-then-authoritative streaming; fast tier serves background work, live chat pinned to reason tier
DONE	Server orchestration, lifecycle, websocket
DONE	Persistence: DB state, memories, state log, proposals
PARTIAL	Remote/auth gates and tunnel modes Auth solid; ngrok launch never captures the public URL (app.py _start_tunnel)

Tier 2: Interaction agents

DONE	Soul seven-subagent loop
DONE	Snapshot-driven focus dispatch + room motion
DONE	Tool loop with bounded rounds and schema gate
DONE	Tool mode control: keyword/smart/always
DONE	Innate tools: memory/journal/status/plan/recall Bounded 7-tool offer; keyword-preferred selection keeps recall/plan reachable
PARTIAL	Vision + computer use Implemented but opt-in; live pyautogui loop untested

Tier 3: Memory system

DONE	Keyword recall (live chat default)
DONE	Semantic backfill pipeline (idempotent)
DONE	Bounded recall candidate scoring
DONE	Embeddings on demand (offline-safe fallbacks)
DONE	Mindpage layer A: pages, writeback, pressure
DONE	Context budget shrink and pressure signal flow Adaptive Mindpage: measures full request, reserves response capacity, commit-safe paging
NOT STARTED	Pagefile KV cache (llama.cpp save/restore)
NOT STARTED	Layer C mmap/memory-tier deep mode

Tier 4: Agency controls

DONE	Constrained pick for tie-break/live/proactive choice
DONE	Deterministic fallback on parse failure
DONE	Proposal-first world edits Safe local tools only behind the gate
DONE	ASK_FIRST enforcement on execution
DONE	Planner step cap + strict payload format Execution proposal-gated; entire-project diagram frames planner as partial, kept as conservative view

Tier 5: Automation

PARTIAL	Routines table + poll loop + routes Create/toggle only; no DELETE route (tests delete via raw SQL)
DONE	Watcher loop + scan-no-content policy
DONE	Built-in routine kinds only (safe, existing functions)
PARTIAL	Consolidation + VACUUM routine Consolidation real; explicit vacuum() hook exists, not yet a scheduled routine kind
PARKED	MCP tool federation

Tier 6: Studio & experience surface

DONE	House HQ 2D + WS sense/event pipeline
DONE	Studio tooling: character sheet + prompt/image route Image render step is ComfyUI-gated
PARTIAL	VRM runtime page + grounding Fully wired on feat/vrm-preview; absent from main
DONE	VCS app pipeline + texture lab integration
PARTIAL	Asset-level 3D model/art iteration Texture/atlas restyle works; geometry shaping is manual in VRoid Studio
PARTIAL	VRoid source pass for base model and outfit corrections Lanyard accessory, true blue clip, base-model QA remain

Tier 7: Voice / periphery / ops

DONE	Identity voice stack (f5/kokoro + defaults)
DONE	Discord bridge and notification logic
DONE	Desktop/app file/vision feature flags Opt-in by design
PARTIAL	Cloud integration (optional: cloud/mirror/deep) Plumbing complete; every cloud path disabled/unconfigured by default
SUPERSEDED	MCP/closed frameworks in new paths

Tier 8: Governance and safety envelope

DONE	Locked spelling and artifact rules (no art upload to Cloudflare) Process-enforced via AGENTS.md; no automated linter
DONE	No default autonomous external side effects Hard-enforced in charter.py / cognition.py
DONE	Deterministic/noise-hardened fallbacks
DONE	Offline honesty and bounded behavior
DONE	Legacy doc posture cleanup + archived stale docs

Verified audit corrections (2026-07-09)

Five concrete defects found by parallel code audit; these downgrade items an earlier revision marked done.

FIXED - Innate tool cap dropped recall. The 7-tool offer is now preference-driven (mind.py _chat_tools_schema): recall/plan/journal tools get guaranteed slots when the message mentions them. Resolved by the 2026-07-09 adaptive Mindpage pass.

OPEN - Routines have no DELETE route. /routines supports list/create/toggle only; tests delete rows via raw SQL. Routines can be disabled but never removed over HTTP.

MOSTLY FIXED - VACUUM. mindpage.py vacuum() now exists as an explicit maintenance hook; not yet a scheduled routine kind, so T5D stays partial.

OPEN - ngrok tunnel is a blind launch. app.py _start_tunnel starts ngrok via bare subprocess.Popen and never captures the public URL; only the cloudflare path goes through preview_mod.ensure.

FIXED - Pressure now drives shrink. The adaptive Mindpage ledger measures the complete LLM request, reserves response capacity, and pages history commit-safely; pressure is grounded telemetry across Soul, prompts, cognition, WS, API, and the House HQ gauge.

Also noted: planner kept as conservative 'partial' in the entire-project diagram (execution proposal-gated, minimal tool set); the fast chat tier serves background/subagent work only since live player chat is pinned to the reason tier; VRM page + grounding is fully wired but exists only on feat/vrm-preview, absent from main.

Next 2-step verification

1. python -m pytest -q tests/test_core.py -q
2. npm.cmd run house:build